MOBSF

ANDROID STATIC ANALYSIS REPORT



Sentinel (1.0.0)

File Name: app-release.apk

Package Name:	com.iheb.sentinel
Scan Date:	Oct. 21, 2026, 11:05 p.m.
App Security Score:	53/100 (MEDIUM RISK)
Grade:	B

FINDINGS SEVERITY

HIGH	MEDIUM	INFO	SECURE	HOTSPOT
2	7	2	2	1

FILE INFORMATION

File Name: app-release.apk
Size: 57.22MB
MD5: cfc37172e6c209030ff8b06e664d573c
SHA1: 4af51c8dcf74b923720b7d4a8c20d8b81a4e8f42
SHA256: e717b6b8c9e36f417254bfa216a92151addabe6e1a282bcb5197797f2600ea53

APP INFORMATION

App Name: Sentinel
Package Name: com.iheb.sentinel
Main Activity: com.iheb.sentinel.MainActivity
Target SDK: 36
Min SDK: 24
Max SDK:
Android Version Name: 1.0.0
Android Version Code: 1

APP COMPONENTS

Activities: 1
Services: 2
Receivers: 2
Providers: 1
Exported Activities: 0

Exported Services: 2
Exported Receivers: 1
Exported Providers: 0

CERTIFICATE INFORMATION

Binary is signed
v1 signature: False
v2 signature: True
v3 signature: False
v4 signature: False
v3.1 signature: False
v3.2 signature: False
X.509 Subject: CN=Android Debug, O=Android, C=US
Signature Algorithm: rsassa_pkcs1v15
Valid From: 2026-10-11 12:29:32+00:00
Valid To: 2056-10-03 12:29:32+00:00
Issuer: CN=Android Debug, O=Android, C=US
Serial Number: 0x1
Hash Algorithm: sha256
md5: 2ae2ac2f14f9935927e6089376f3416a
sha1: 976e2b0bcedf01e8ab60cad7eb8f642cb7276243
sha256: c87e20d9c7b4950a046217f028739d630d38a11d76bbbbb725cd7b2b682ef6d9
sha512: a6bbfe8f0d9ccd29bee8a4ddab1d56c7fe3770126e071ea3b544035d0b18cd405b7bb42f621a84ad162b23bd423ed64f7e60fc0bbeb9eed839cc26a97fe0ad57
PublicKey Algorithm: rsa
Bit Size: 2048
Fingerprint: 8456fe7d3ab06537d1ad765edc5db074ef126124c99e76e5d3e93c2ccd2c3c46
Found 1 unique certificates

APPLICATION PERMISSIONS

PERMISSION	STATUS	INFO	DESCRIPTION
android.permission.USE_BIOMETRIC	normal	allows use of device-supported biometric modalities.	Allows an app to use device supported biometric modalities.

PERMISSION	STATUS	INFO	DESCRIPTION
android.permission.QUERY_ALL_PACKAGES	normal	enables querying any normal app on the device.	Allows query of any normal app on the device, regardless of manifest declarations.
android.permission.REQUEST_DELETE_PACKAGES	normal	enables an app to request package deletions.	Allows an application to request deleting packages.
android.permission.POST_NOTIFICATIONS	dangerous	allows an app to post notifications.	Allows an app to post notifications
android.permission.ACCESS_NETWORK_STATE	normal	view network status	Allows an application to view the status of all networks.
android.permission.INTERNET	normal	full Internet access	Allows an application to create network sockets.
android.permission.FOREGROUND_SERVICE	normal	enables regular apps to use Service.startForeground.	Allows a regular application to use Service.startForeground.
android.permission.FOREGROUND_SERVICE_SYSTEM_EXEMPTED	normal	allows system-exempted types of foreground services.	Allows a regular application to use Service.startForeground with the type "systemExempted". Apps are allowed to use this type only in the use cases listed in ServiceInfo.FOREGROUND_SERVICE_TYPE_SYSTEM_EXEMPTED .
android.permission.SCHEDULE_EXACT_ALARM	normal	permits exact alarm scheduling for background work.	Allows an app to use exact alarm scheduling APIs to perform timing sensitive background work.
com.iheb.sentinel.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION	unknown	Unknown permission	Unknown permission from android reference

APKID ANALYSIS

<table>
<tr><th>FILE</th><th>DETAILS</th></tr>
<tr><td>cfc37172e6c209030ff8b06e664d573c.apk</td><td>
<table>
<tr><th>FINDINGS</th><th>DETAILS</th></tr>
<tr><td>Manipulator Found</td><td>Resources Confusion</td></tr>
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</td></tr>
<tr><td>classes.dex</td><td>
<table>
<tr><th>FINDINGS</th><th>DETAILS</th></tr>
<tr><td>Anti-VM Code</td><td>Build.TAGS check</td></tr>
<tr><td>Compiler</td><td>r8</td></tr>
</table>
</td></tr>
</table>

NETWORK SECURITY

NO	SCOPE	SEVERITY	DESCRIPTION

CERTIFICATE ANALYSIS

HIGH: 1 | WARNING: 0 | INFO: 1

TITLE	SEVERITY	DESCRIPTION
Signed Application	info	Application is signed with a code signing certificate
Application signed with debug certificate	high	Application signed with a debug certificate. Production application must not be shipped with a debug certificate.

MANIFEST ANALYSIS

HIGH: 1 | WARNING: 3 | INFO: 0 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	DESCRIPTION
1	App can be installed on a vulnerable unpatched Android version 7.0, [minSdk=24]	high	This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.
2	Service (com.iheb.sentinel.SentinelNotificationListener) is Protected by a permission, but the protection level of the permission should be checked. Permission: android.permission.BIND_NOTIFICATION_LISTENER_SERVICE [android:exported=true]	warning	A Service is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.
3	Service (com.iheb.sentinel.firewall.SentinelVpnService) is Protected by a permission, but the protection level of the permission should be checked. Permission: android.permission.BIND_VPN_SERVICE [android:exported=true]	warning	A Service is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.
4	Broadcast Receiver (androidx.profileinstaller.ProfileInstallReceiver) is Protected by a permission, but the protection level of the permission should be checked. Permission: android.permission.DUMP [android:exported=true]	warning	A Broadcast Receiver is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. It is protected by a permission which is not defined in the analysed application. As a result, the protection level of the permission should be checked where it is defined. If it is set to normal or dangerous, a malicious application can request and obtain the permission and interact with the component. If it is set to signature, only applications signed with the same certificate can obtain the permission.

CODE ANALYSIS

HIGH: 0 | WARNING: 3 | INFO: 2 | SECURE: 1 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	STANDARDS	FILES
1	SHA-1 is a weak hash known to have hash collisions.	warning	CWE: CWE-327: Use of a Broken or Risky Cryptographic Algorithm OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-4	defpackage/vb.java

NO	ISSUE	SEVERITY	STANDARDS	FILES
2	The App logs information. Sensitive information should never be logged.	info	CWE: CWE-532: Insertion of Sensitive Information into Log File OWASP MASVS: MSTG-STORAGE-3	com/iheb/sentinel/SentinelNotificationListener.java defpackage/a1.java defpackage/b1.java defpackage/b7.java defpackage/c7.java defpackage/ca.java defpackage/cm.java defpackage/di.java defpackage/e5.java defpackage/e9.java defpackage/f0.java defpackage/g7.java defpackage/gm.java defpackage/il.java defpackage/j5.java defpackage/l5.java defpackage/m0.java defpackage/m9.java defpackage/me.java defpackage/ml.java defpackage/n8.java defpackage/o1.java defpackage/o7.java defpackage/o8.java defpackage/pc.java defpackage/r8.java defpackage/s1.java defpackage/t9.java defpackage/v9.java defpackage/vd.java defpackage/w5.java defpackage/y0.java defpackage/y7.java defpackage/yb.java defpackage/zl.java io/flutter/embedding/engine/FlutterJNI.java io/flutter/embedding/engine/renderer/FlutterRenderer$ImageTextureRegistryEntry.java io/flutter/embedding/engine/renderer/d.java io/flutter/plugin/editing/b.java

NO	ISSUE	SEVERITY	STANDARDS	FILES
				io/flutter/plugins/GeneratedPluginRegistrant.java io/flutter/view/AccessibilityViewEmbedder.java
3	This App copies data to clipboard. Sensitive data should not be copied to clipboard as other applications can access it.	info	OWASP MASVS: MSTG-STORAGE-10	defpackage/f0.java defpackage/sb.java defpackage/y7.java
4	The App uses an insecure Random Number Generator.	warning	CWE: CWE-330: Use of Insufficiently Random Values OWASP Top 10: M5: Insufficient Cryptography OWASP MASVS: MSTG-CRYPTO-6	defpackage/b8.java defpackage/h.java defpackage/kf.java defpackage/p7.java defpackage/q7.java
5	This App may have root detection capabilities.	secure	OWASP MASVS: MSTG-RESILIENCE-1	defpackage/a1.java
6	IP Address disclosure	warning	CWE: CWE-200: Information Exposure OWASP MASVS: MSTG-CODE-2	com/iheb/sentinel/firewall/SentinelVpnService.java defpackage/sd.java

SHARED LIBRARY BINARY ANALYSIS

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
1	armeabi-v7a/libflutter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Not Applicable info RELRO checks are not applicable for Flutter/Dart binaries	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__vsnprintf_chk', '__read_chk', '__memcpy_chk', '__strcpy_chk', '__strlen_chk', '__memmove_chk', '__vsprintf_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
2	armeabi-v7a/libdartjni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
3	armeabi-v7a/libapp.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	No RELRO high This shared object does not have RELRO enabled. The entire GOT (.got and .got.plt both) are writable. Without this compiler flag, buffer overflows on a global variable can overwrite GOT entries. Use the option -z,relro,-z,now to enable full RELRO and only -z,relro to enable partial RELRO.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
4	arm64-v8a/libflutter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Not Applicable info RELRO checks are not applicable for Flutter/Dart binaries	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__fwrite_chk', '__read_chk', '__write_chk', '__memcpy_chk', '__strcpy_chk', '__pread_chk', '__strlen_chk', '__memmove_chk', '__poll_chk', '__readlink_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
5	arm64-v8a/libdartjni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
6	arm64-v8a/libapp.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	No RELRO high This shared object does not have RELRO enabled. The entire GOT (.got and .got.plt both) are writable. Without this compiler flag, buffer overflows on a global variable can overwrite GOT entries. Use the option -z,relro,-z,now to enable full RELRO and only -z,relro to enable partial RELRO.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
7	x86_64/libflutter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Not Applicable info RELRO checks are not applicable for Flutter/Dart binaries	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__fwrite_chk', '__read_chk', '__write_chk', '__memcpy_chk', '__strcpy_chk', '__pread_chk', '__strlen_chk', '__memmove_chk', '__poll_chk', '__readlink_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
8	x86_64/libdartjni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
9	x86_64/libapp.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	No RELRO high This shared object does not have RELRO enabled. The entire GOT (.got and .got.plt both) are writable. Without this compiler flag, buffer overflows on a global variable can overwrite GOT entries. Use the option -z,relro,-z,now to enable full RELRO and only -z,relro to enable partial RELRO.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
10	armeabi-v7a/libflutter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Not Applicable info RELRO checks are not applicable for Flutter/Dart binaries	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__vsnprintf_chk', '__read_chk', '__memcpy_chk', '__strcpy_chk', '__strlen_chk', '__memmove_chk', '__vsprintf_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
11	armeabi-v7a/libdartjni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
12	armeabi-v7a/libapp.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	No RELRO high This shared object does not have RELRO enabled. The entire GOT (.got and .got.plt both) are writable. Without this compiler flag, buffer overflows on a global variable can overwrite GOT entries. Use the option -z,relro,-z,now to enable full RELRO and only -z,relro to enable partial RELRO.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
13	arm64-v8a/libflutter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Not Applicable info RELRO checks are not applicable for Flutter/Dart binaries	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__fwrite_chk', '__read_chk', '__write_chk', '__memcpy_chk', '__strcpy_chk', '__pread_chk', '__strlen_chk', '__memmove_chk', '__poll_chk', '__readlink_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
14	arm64-v8a/libdartjni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
15	arm64-v8a/libapp.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	No RELRO high This shared object does not have RELRO enabled. The entire GOT (.got and .got.plt both) are writable. Without this compiler flag, buffer overflows on a global variable can overwrite GOT entries. Use the option -z,relro,-z,now to enable full RELRO and only -z,relro to enable partial RELRO.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
16	x86_64/libflutter.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Not Applicable info RELRO checks are not applicable for Flutter/Dart binaries	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	True info The binary has the following fortified functions: ['__fwrite_chk', '__read_chk', '__write_chk', '__memcpy_chk', '__strcpy_chk', '__pread_chk', '__strlen_chk', '__memmove_chk', '__poll_chk', '__readlink_chk']	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
17	x86_64/libdartjni.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	True info This binary has a stack canary value added to the stack so that it will be overwritten by a stack buffer that overflows the return address. This allows detection of overflows by verifying the integrity of the canary before function return.	Full RELRO info This shared object has full RELRO enabled. RELRO ensures that the GOT cannot be overwritten in vulnerable ELF binaries. In Full RELRO, the entire GOT (.got and .got.plt both) is marked as read-only.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NO	SHARED OBJECT	NX	PIE	STACK CANARY	RELRO	RPATH	RUNPATH	FORTIFY	SYMBOLS STRIPPED
18	x86_64/libapp.so	True info The binary has NX bit set. This marks a memory page non-executable making attacker injected shellcode non-executable.	Dynamic Shared Object (DSO) info The shared object is build with -fPIC flag which enables Position independent code. This makes Return Oriented Programming (ROP) attacks much more difficult to execute reliably.	False high This binary does not have a stack canary value added to the stack. Stack canaries are used to detect and prevent exploits from overwriting return address. Use the option -fstack-protector-all to enable stack canaries. Not applicable for Dart/Flutter libraries unless Dart FFI is used.	No RELRO high This shared object does not have RELRO enabled. The entire GOT (.got and .got.plt both) are writable. Without this compiler flag, buffer overflows on a global variable can overwrite GOT entries. Use the option -z,relro,-z,now to enable full RELRO and only -z,relro to enable partial RELRO.	None info The binary does not have run-time search path or RPATH set.	None info The binary does not have RUNPATH set.	False warning The binary does not have any fortified functions. Fortified functions provides buffer overflow checks against glibc's commons insecure functions like strcpy, gets etc. Use the compiler option -D_FORTIFY_SOURCE=2 to fortify functions. This check is not applicable for Dart/Flutter libraries.	True info Symbols are stripped.

NIAP ANALYSIS v1.3

NO	IDENTIFIER	REQUIREMENT	FEATURE	DESCRIPTION

BEHAVIOUR ANALYSIS

RULE ID	BEHAVIOUR	LABEL	FILES
00035	Query the list of the installed packages	reflection	defpackage/vb.java
00022	Open a file from given absolute path of the file	file	defpackage/k9.java io/flutter/embedding/engine/FlutterJNI.java
00013	Read file and put it into a stream	file	defpackage/gm.java defpackage/h8.java defpackage/hg.java defpackage/l5.java defpackage/mh.java defpackage/w5.java defpackage/yb.java
00209	Get pixels from the latest rendered image	collection	defpackage/e9.java
00210	Copy pixels from the latest rendered image into a Bitmap	collection	defpackage/e9.java
00004	Get filename and put it to JSON object	file collection	defpackage/g2.java defpackage/l5.java defpackage/y0.java
00014	Read file into a stream and put it into a JSON object	file	defpackage/gm.java defpackage/l5.java

RULE ID	BEHAVIOUR	LABEL	FILES
00161	Perform accessibility service action on accessibility node info	accessibility service	io/flutter/view/AccessibilityViewEmbedder.java io/flutter/view/b.java
00173	Get bounds in screen of an AccessibilityNodeInfo and perform action	accessibility service	io/flutter/view/AccessibilityViewEmbedder.java
00063	Implicit intent(view a web page, make a phone call, etc.)	control	com/iheb/sentinel/MainActivity.java defpackage/o8.java
00091	Retrieve data from broadcast	collection	defpackage/o8.java
00036	Get resource file from res/raw directory	reflection	defpackage/o8.java
00125	Check if the given file path exist	file	defpackage/l5.java

ABUSED PERMISSIONS

TYPE	MATCHES	PERMISSIONS
Malware Permissions	2/25	android.permission.ACCESS_NETWORK_STATE, android.permission.INTERNET
Other Common Permissions	1/44	android.permission.FOREGROUND_SERVICE

Malware Permissions:
Top permissions that are widely abused by known malware.
Other Common Permissions:
Permissions that are commonly abused by known malware.

OFAC SANCTIONED COUNTRIES

This app may communicate with the following OFAC sanctioned list of countries.

DOMAIN	COUNTRY/REGION

DOMAIN MALWARE CHECK

DOMAIN	STATUS	GEOLOCATION
api.pwnedpasswords.com	ok	**IP:** 104.17.175.230 **Country:** United States of America **Region:** California **City:** San Francisco **Latitude:** 37.775700 **Longitude:** -122.395203 **View:** Google Map
developer.android.com	ok	**IP:** 142.251.209.110 **Country:** United States of America **Region:** California **City:** Mountain View **Latitude:** 37.405991 **Longitude:** -122.078514 **View:** Google Map
stopstalkerware.org	ok	**IP:** 204.197.251.106 **Country:** United States of America **Region:** Georgia **City:** Atlanta **Latitude:** 33.749001 **Longitude:** -84.387978 **View:** Google Map

DOMAIN	STATUS	GEOLOCATION
api.flutter.dev	ok	**IP:** 199.36.158.100 **Country:** United States of America **Region:** California **City:** Mountain View **Latitude:** 37.405991 **Longitude:** -122.078514 **View:** Google Map
www.w3.org	ok	**IP:** 104.18.23.19 **Country:** United States of America **Region:** California **City:** San Francisco **Latitude:** 37.775700 **Longitude:** -122.395203 **View:** Google Map
fonts.gstatic.com	ok	**IP:** 142.251.209.99 **Country:** United States of America **Region:** California **City:** Mountain View **Latitude:** 37.405991 **Longitude:** -122.078514 **View:** Google Map
dartbug.com	ok	**IP:** 216.239.38.21 **Country:** United States of America **Region:** California **City:** Mountain View **Latitude:** 37.405991 **Longitude:** -122.078514 **View:** Google Map
ns.adobe.com	ok	No Geolocation information available.

DOMAIN	STATUS	GEOLOCATION
github.com	ok	**IP:** 140.82.121.4 **Country:** United States of America **Region:** California **City:** San Francisco **Latitude:** 37.775700 **Longitude:** -122.395203 **View:** Google Map
docs.flutter.dev	ok	**IP:** 199.36.158.100 **Country:** United States of America **Region:** California **City:** Mountain View **Latitude:** 37.405991 **Longitude:** -122.078514 **View:** Google Map
www.unicode.org	ok	**IP:** 104.26.11.47 **Country:** United States of America **Region:** California **City:** San Francisco **Latitude:** 37.775700 **Longitude:** -122.395203 **View:** Google Map

EMAILS

EMAIL	FILE
_httpparser@17463476.responsepa _timer@1026248._internal _colorfilter@35065589.srgbtoline _headervalue@17463476.parse _growablelist@0150898._ofother _future@5048458.immediate	

EMAIL	FILE
_growablelist@0150898._literal _file@16069316.fromrawpat _compressednode@308137193.single _list@0150898.of _growablelist@0150898._literal8 _typeerror@0150898._create _growablelist@0150898._ofarray _imagefilter@35065589.composed _nativesocket@16069316.normal _assertionerror@0150898._create _list@0150898.generate _timer@1026248.periodic _list@0150898._ofefficie _list@0150898.empty _uri@0150898.file _hashcollisionnode@308137193.fromcollis _growablelist@0150898._literal5 _list@0150898._ofgrowabl authenticationscheme@17463476.fromstring _uri@0150898.directory _future@5048458.immediatee _imagefilter@35065589.fromcolorf _growablelist@0150898._ofefficie _growablelist@0150898._literal7 _double@0150898.fromintege _bytebuffer@8027147._new _growablelist@0150898.of _uri@0150898.notsimple _list@0150898._ofarray _assetmanifestbin@325287047.fromstanda _growablelist@0150898.withcapaci _receiveportimpl@1026248.fromrawrec _colorfilter@35065589.lineartosr ngstreamsubscription@5048458.zoned _semanticsgeometry@27266271.root _growablelist@0150898._literal3 _growablelist@0150898._literal4 _invocationmirror@0150898._withtype _growablelist@0150898._literal6 _growablelist@0150898._ofgrowabl _link@16069316.fromrawpat _growablelist@0150898._literal2 storationinformation@39124995.fromserial _list@0150898._ofother	lib/armeabi-v7a/libapp.so

EMAIL	FILE
_directory@16069316.fromrawpat _growablelist@0150898._literal1 _growablelist@0150898.generate _imagefilter@35065589.blur	
appro@openssl.org	lib/arm64-v8a/libflutter.so
appro@openssl.org	lib/x86_64/libflutter.so
_httpparser@17463476.responsepa _timer@1026248._internal _colorfilter@35065589.srgbtoline _headervalue@17463476.parse _growablelist@0150898._ofother _future@5048458.immediate _growablelist@0150898._literal _file@16069316.fromrawpat _compressednode@308137193.single _list@0150898.of _growablelist@0150898._literal8 _typeerror@0150898._create _growablelist@0150898._ofarray _imagefilter@35065589.composed _nativesocket@16069316.normal _assertionerror@0150898._create _list@0150898.generate _timer@1026248.periodic _list@0150898._ofefficie _list@0150898.empty _uri@0150898.file _hashcollisionnode@308137193.fromcollis _growablelist@0150898._literal5 _list@0150898._ofgrowabl authenticationscheme@17463476.fromstring _uri@0150898.directory _future@5048458.immediatee _imagefilter@35065589.fromcolorf _growablelist@0150898._ofefficie _growablelist@0150898._literal7 _double@0150898.fromintege _bytebuffer@8027147._new _growablelist@0150898.of _uri@0150898.notsimple	apktool_out/lib/armeabi-v7a/libapp.so

EMAIL	FILE
_list@0150898._ofarray _manifestbin@325287047.fromstanda _growablelist@0150898.withcapaci	
_receiveportimpl@1026248.fromrawrec _colorfilter@35065589.lineartosr ngstreamsubscription@5048458.zoned _semanticsgeometry@27266271.root _growablelist@0150898._literal3 _growablelist@0150898._literal4 _invocationmirror@0150898._withtype _growablelist@0150898._literal6 _growablelist@0150898._ofgrowabl _link@16069316.fromrawpat _growablelist@0150898._literal2 storationinformation@39124995.fromserial _list@0150898._ofother _directory@16069316.fromrawpat _growablelist@0150898._literal1 _growablelist@0150898.generate _imagefilter@35065589.blur	
appro@openssl.org	apktool_out/lib/arm64-v8a/libflutter.so
appro@openssl.org	apktool_out/lib/x86_64/libflutter.so

HARDCODED SECRETS

POSSIBLE SECRETS
000000010010001101000101011001111000100110101011110011011110111100010203040506071011121314151617202122232425262730313233343536374041424344454647505152535455565760616263646566677071727374757677000102030405060708090a0b0c0d0e0f101112131415161718191a1b1c1d1e1f202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f404142434445464748494a4b4c4d4e4f505152535455565758595a5b5c5d5e5f606162636465666768696a6b6c6d6e6f707172737475767778797a7b7c7d7e7f808182838485868788898a8b8c8d8e8f909192939495969798999a9b9c9d9e9fa0a1a2a3a4a5a6a7a8a9aaabacadaeafb0b1b2b3b4b5b6b7b8b9babbbcbdbebfc0c1c2c3c4c5c6c7c8c9cacbcccdcecfd0d1d2d3d4d5d6d7d8d9dadbdcdddedfe0e1e2e3e4e5e6e7e8e9eaebecedeeeff0f1f2f3f4f5f6f7f8f9fafbfcfdfeff
000102030405060708091011121314151617181920212223242526272829303132333435363738394041424344454647484950515253545556575859606162636465666768697071727374757677787980818283848586878889909192939495969798999

POSSIBLE SECRETS
000102030405060708091011121314151617181920212223242526272829303132333435363738394041424344454647484950515253545556575859606162636465666768697071727374757677787980818283848586878889909192939495969798999000000010010001101000101011001111000100110101011110011011110111100010203040506071011121314151617202122232425262730313233343536374041424344454647505152535455565760616263646566677071727374757677000102030405060708090a0b0c0d0e0f10111213141516171819 1a1b1c1d1e1f202122232425262728292a2b2c2d2e2f303132333435363738393a3b3c3d3e3f404142434445464748494a4b4c4d4e4f505152535455565758595a5b5c5d5e5f606162636465666768696a6b6c6d6e6f707172737475767778797a7b7c7d7e7f808182838485868788898a8b8c8d8e8f909192939495969798999a9b9c9d9e9fa0a1a2a3a4a5a6a7a8a9aaabacadaeafb0b1b2b3b4b5b6b7b8b9babbbcbdbebfc0c1c2c3c4c5c6c7c8c9cacbcccdcecfd0d1d2d3d4d5d6d7d8d9dadbdcdddedfe0e1e2e3e4e5e6e7e8e9eaebecedeeeff0f1f2f3f4f5f6f7f8f9fafbfcfdfeff
0123456789abcdefABCDEF
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54840850d336e22dca0bd8224a6dc294721e62642eab203f8acbe35a7ea402b8
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711c43d85e02884b5bc74994b87fab1fe84c9f9548495a1c2f0b779b7995f861
8df24be66531469e576a806749a0202ae26b8d08
8f279458dd6c19801c903d9e471c8564d5b8cc0b639e5bff1cb2c9a1f0f2a647
d01db080f4e540c5eb64f567df67d82e951bd72b739e26ccbe55fc9c1ea03f4c
deb287481e3ce9468f3434937ced4240a70539ca
ed47cb984d1f274788ab1d7e5e435d25e9facc7adbdf87168f71dda4b1d24b2c
vkCmdPushDescriptorSetWithTemplate2KHR

SCAN LOGS

Timestamp	Event	Error
2026-10-21 23:05:41	Generating Hashes	OK
2026-10-21 23:05:41	Extracting APK	OK
2026-10-21 23:05:41	Unzipping	OK
2026-10-21 23:05:42	Parsing APK with androguard	OK
2026-10-21 23:05:42	Extracting APK features using aapt/aapt2	OK
2026-10-21 23:05:42	Getting Hardcoded Certificates/Keystores	OK
2026-10-21 23:05:43	Parsing AndroidManifest.xml	OK
2026-10-21 23:05:44	Extracting Manifest Data	OK
2026-10-21 23:05:44	Manifest Analysis Started	OK
2026-10-21 23:05:44	Performing Static Analysis on: Sentinel (com.iheb.sentinel)	OK

2026-10-21 23:05:46	Fetching Details from Play Store: com.iheb.sentinel	OK
2026-10-21 23:05:48	Checking for Malware Permissions	OK
2026-10-21 23:05:48	Fetching icon path	OK
2026-10-21 23:05:48	Library Binary Analysis Started	OK
2026-10-21 23:05:48	Analyzing lib/armeabi-v7a/libflutter.so	OK
2026-10-21 23:05:48	Analyzing lib/armeabi-v7a/libdartjni.so	OK
2026-10-21 23:05:48	Analyzing lib/armeabi-v7a/libapp.so	OK
2026-10-21 23:05:48	Analyzing lib/arm64-v8a/libflutter.so	OK
2026-10-21 23:05:48	Analyzing lib/arm64-v8a/libdartjni.so	OK
2026-10-21 23:05:48	Analyzing lib/arm64-v8a/libapp.so	OK
2026-10-21 23:05:48	Analyzing lib/x86_64/libflutter.so	OK
2026-10-21 23:05:52	Analyzing lib/x86_64/libdartjni.so	OK

2026-10-21 23:05:52	Analyzing lib/x86_64/libapp.so	OK
2026-10-21 23:05:52	Analyzing apktool_out/lib/armeabi-v7a/libflutter.so	OK
2026-10-21 23:05:52	Analyzing apktool_out/lib/armeabi-v7a/libdartjni.so	OK
2026-10-21 23:05:52	Analyzing apktool_out/lib/armeabi-v7a/libapp.so	OK
2026-10-21 23:05:53	Analyzing apktool_out/lib/arm64-v8a/libflutter.so	OK
2026-10-21 23:05:53	Analyzing apktool_out/lib/arm64-v8a/libdartjni.so	OK
2026-10-21 23:05:53	Analyzing apktool_out/lib/arm64-v8a/libapp.so	OK
2026-10-21 23:05:53	Analyzing apktool_out/lib/x86_64/libflutter.so	OK
2026-10-21 23:05:53	Analyzing apktool_out/lib/x86_64/libdartjni.so	OK
2026-10-21 23:05:53	Analyzing apktool_out/lib/x86_64/libapp.so	OK
2026-10-21 23:05:54	Reading Code Signing Certificate	OK

2026-10-21 23:05:54	Running APKiD 3.1.0	OK
2026-10-21 23:05:58	Detecting Trackers	OK
2026-10-21 23:05:58	Decompiling APK to Java with JADX	OK
2026-10-21 23:06:00	Converting DEX to Smali	OK
2026-10-21 23:06:00	Code Analysis Started on - java_source	OK
2026-10-21 23:06:01	Android SBOM Analysis Completed	OK
2026-10-21 23:06:01	Android SAST Completed	OK
2026-10-21 23:06:01	Android API Analysis Started	OK
2026-10-21 23:06:02	Android API Analysis Completed	OK
2026-10-21 23:06:02	Android Permission Mapping Started	OK
2026-10-21 23:06:02	Android Permission Mapping Completed	OK
2026-10-21 23:06:02	Android Behaviour Analysis Started	OK

2026-10-21 23:06:03	Android Behaviour Analysis Completed	OK
2026-10-21 23:06:03	Extracting Emails and URLs from Source Code	OK
2026-10-21 23:06:03	Email and URL Extraction Completed	OK
2026-10-21 23:06:04	Extracting String data from APK	OK
2026-10-21 23:06:04	Extracting String data from SO	OK
2026-10-21 23:06:06	Extracting String data from Code	OK
2026-10-21 23:06:06	Extracting String values and entropies from Code	OK
2026-10-21 23:06:06	Performing Malware check on extracted domains	OK
2026-10-21 23:07:05	Saving to Database	OK

Report Generated by - MobSF v4.5.3

Mobile Security Framework (MobSF) is an automated, all-in-one mobile application (Android/iOS/Windows) pen-testing, malware analysis and security assessment framework capable of performing static and dynamic analysis.

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